Non-Profit Quiz /Tutor Bot

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI-15

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Academic Year: 2026

Problem Statement & Objectives

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2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features
2. Overall Architecture / Workflow
3. Tools & Technologies Used

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

1. Problem Statement and Objective

1.1. Problem Statement

The non-profit sector requires effective handling of donor communications, but learners often lack practical exposure and meaningful feedback during training. Existing methods are mostly theoretical and fail to develop real-world decision-making skills.

There is a need for an AI-driven system that simulates donor interaction scenarios, evaluates user responses using Large Language Models, and provides contextual feedback to enhance learning and bridge the gap between theory and practice.

1.2. Project Objective

The primary objective of this project is to develop an AI-powered Non-Profit Quiz/Tutor Bot that enhances learning through interactive and intelligent assessment. The system aims to leverage Generative AI and Large Language Models (LLMs) to create a personalized educational experience focused on the non-profit domain.

Key objectives include:

- To design and develop an interactive quiz-based learning platform for non-profit concepts
- To generate context-aware quiz questions from donor email datasets using AI
- To evaluate user responses intelligently using LLMs instead of simple right/wrong logic
- To provide detailed, contextual explanations for incorrect answers to improve understanding
- To implement Retrieval-Augmented Generation (RAG) for accurate and relevant knowledge retrieval
- To build a scalable and modular system architecture using Python, vector databases, and APIs
- To enhance user engagement through a chatbot-based interface

1.2. Scope of the project

The purpose of this document is to describe the design, architecture, and functionality of the Generative AI-based Non-Profit Quiz /Tutor Bot. This document outlines the overall system structure, application workflow, data flow, and technical components used to build the solution.

The system is designed to assist users in learning about the non-profit sector through an interactive quiz-based educational platform powered by Generative Artificial Intelligence and Large Language Models (LLMs). The bot evaluates user responses, provides contextual explanations for incorrect answers, and helps reinforce learning through intelligent feedback.

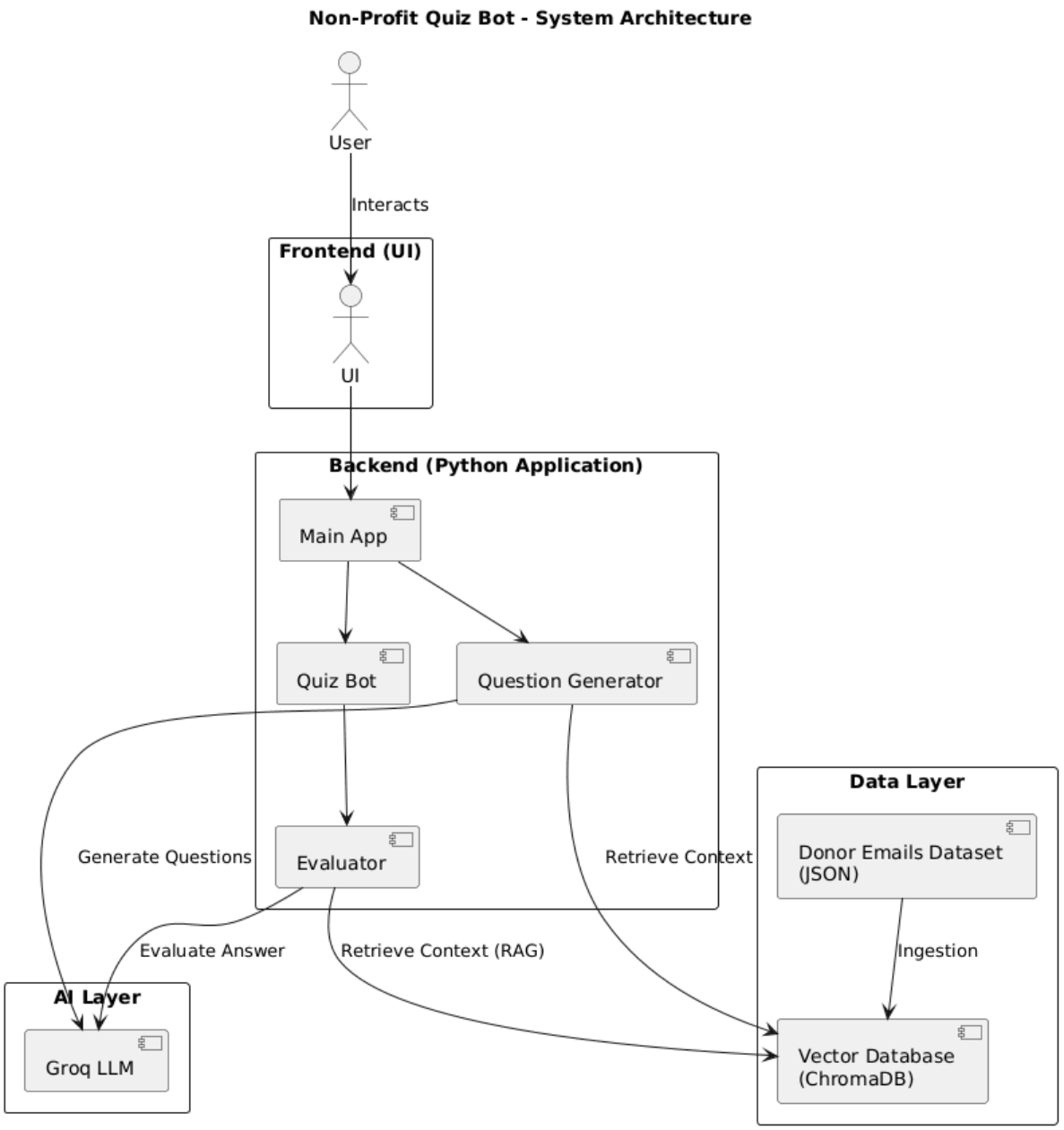
This document covers the system design, process flow, component architecture, API integrations, and design considerations required to implement the application. It serves as a technical reference for developers, researchers, and stakeholders involved in the development and maintenance of the system.

2. Proposed Solution

2.1. Key Features

- AI-Powered Quiz & Evaluation System
- Retrieval-Augmented Generation (RAG) with Vector Database
- AI Non-Profit Advisor & Context-Aware Q&A
- Personalized Learning & Feedback System
- Semantic Search with Embeddings
- Scenario-Based Non-Profit Learning
- Low-Latency LLM Inference (Groq + LLaMA 3.1)
- Modular & Scalable Architecture
- Web-Based Interactive Interface
- Efficient Vector Store Caching

2.2. Overall Architecture/Workflow



2.2.1. User Interface (Frontend Layer)

The Frontend Layer acts as the interaction point between the user and the system. Users access the application through a chatbot-style interface where they can answer quiz questions and receive explanations.

The interface is designed to be simple and interactive so that users can easily engage with the learning system. It sends user responses to the backend for processing and displays the feedback generated by the AI system. Technologies such as Streamlit or web-based UI frameworks can be used to build this layer.

The main responsibilities of the frontend include:

- Accepting user responses to quiz questions
- Displaying quiz questions generated by the system
- Showing evaluation results and AI-generated explanations
- Managing user interaction with the chatbot interface

2.2.2. Backend Layer (Python Application)

The Backend Layer is the core processing unit of the application. It manages the logic required for generating quiz questions, evaluating answers, and retrieving contextual knowledge from the database.

The backend consists of several modules:

- **Main Application Module:** The main application module controls the overall workflow of the system. It manages communication between the frontend, quiz engine, evaluation system, AI model, and database.
- **Quiz Bot Module:** The Quiz Bot module manages quiz interactions. It sends quiz questions to the user, collects answers, and forwards them for evaluation.
- **Question Generator:** The evaluator analyzes the user's answers and determines whether they are correct or incorrect. It sends the response to the AI model for deeper analysis and explanation generation.

2.2.3. AI Layer

The AI Layer is responsible for intelligent reasoning and explanation generation. This layer integrates a Large Language Model (LLM) Groq API - LLaMA 3.1 to perform natural language understanding and generate contextual responses.

The AI model performs the following functions:

- Evaluates the correctness of user answers
- Generates explanations for incorrect answers
- Provides contextual feedback to enhance learning
- Uses retrieved knowledge to ensure accurate responses

2.2.4. Data Layer

The Data Layer manages the storage and retrieval of knowledge used by the system. It contains structured and processed information related to non-profit sector topics.

The main components of this layer include:

- **Donor Emails Dataset (JSON):** The dataset contains donor communication examples and non-profit related content stored in JSON format. This dataset is used to train and support the quiz generation process.
- **Vector Database (ChromaDB):** The dataset is converted into embeddings and stored in a vector database. This allows the system to perform semantic search and retrieve relevant context when evaluating answers or generating questions.

2.2.5. System Workflow

The interaction between different components follows

The interaction between different components follows a structured workflow:

1. The user interacts with the frontend interface and submits an answer.

2. The frontend sends the response to the backend main application.

3. The Quiz Bot module manages the quiz session and forwards the answer to the Evaluator module.

4. The evaluator retrieves relevant context from the Vector Database (ChromaDB) using semantic search.

5. The retrieved information is sent to the Groq LLM in the AI layer.

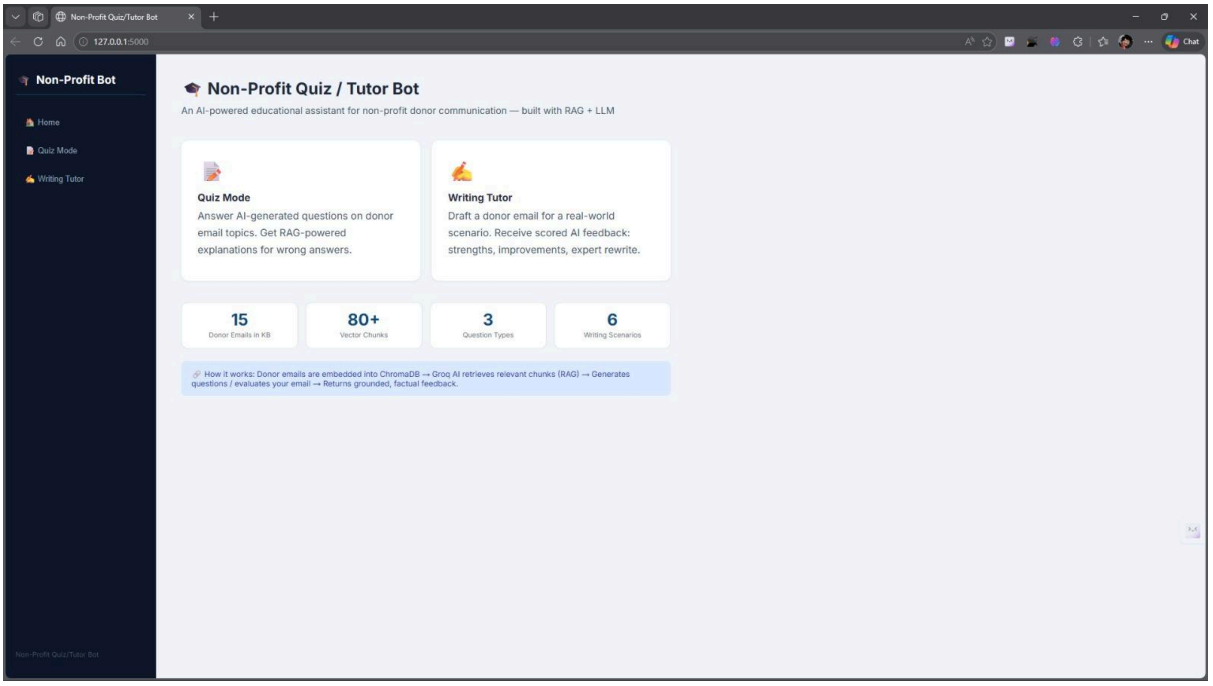
6. The AI model evaluates the answer and generates an explanation.
7. The result is returned to the backend and displayed to the user interface.

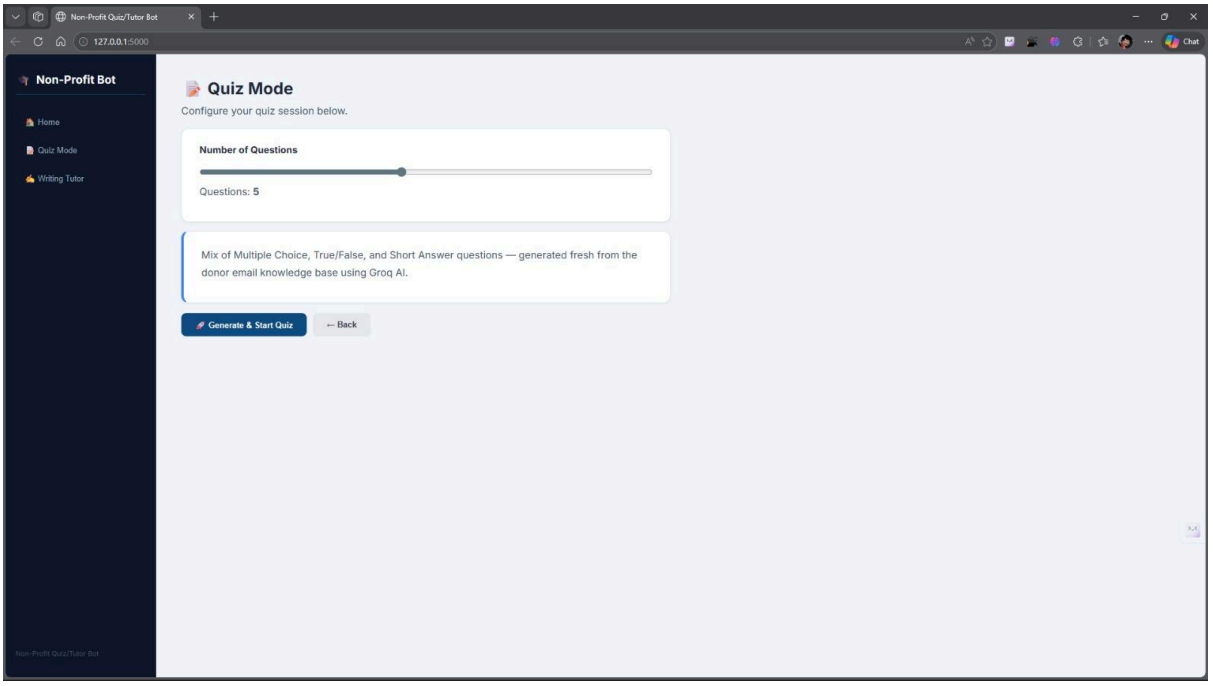
2.3. Tools & Technologies Used

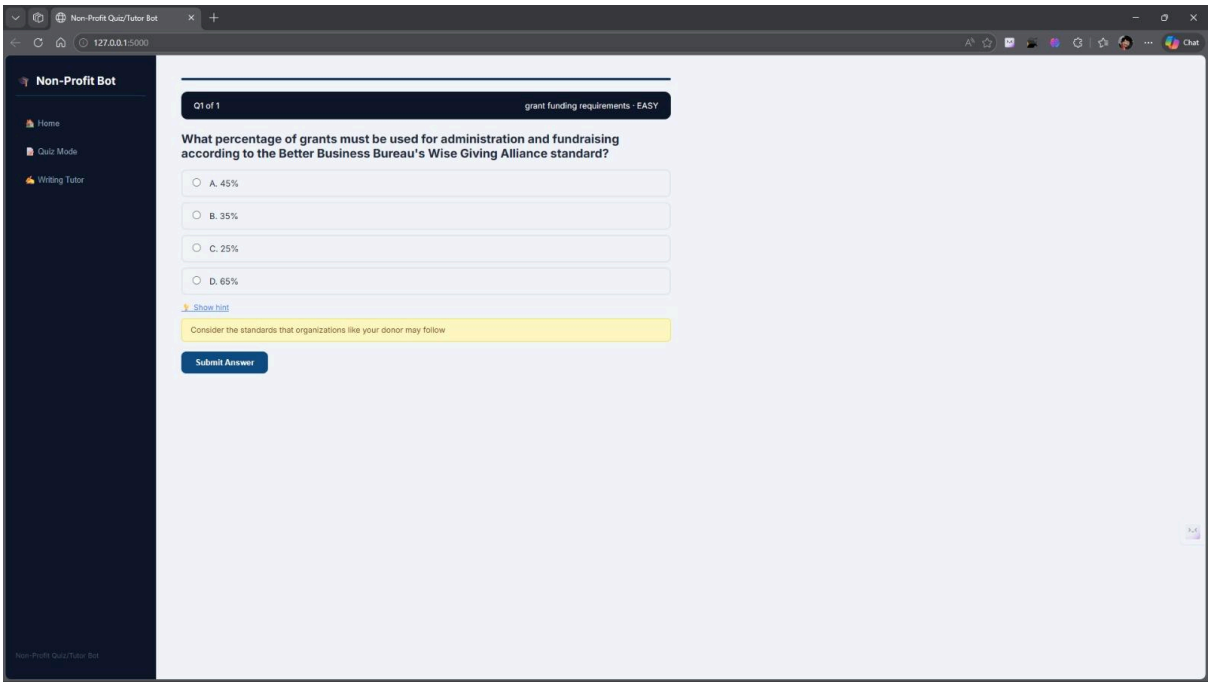
- **Programming Language:** Python
- **Frontend Technologies:** HTML, CSS, JavaScript / Streamlit
- **Backend Framework:** Python-based application modules
- **Large Language Model (LLM):** LLaMA 3.1 (via Groq API)
- **API Integration:** Groq API
- **Vector Database:** ChromaDB
- **Embedding Techniques:** Sentence Transformers / Embedding Models
- **Framework/Libraries:** LangChain
- **Data Format:** JSON (Donor Email Dataset)
- **Architecture Approach:** Retrieval-Augmented Generation (RAG)
- **Development Tools:** VS Code

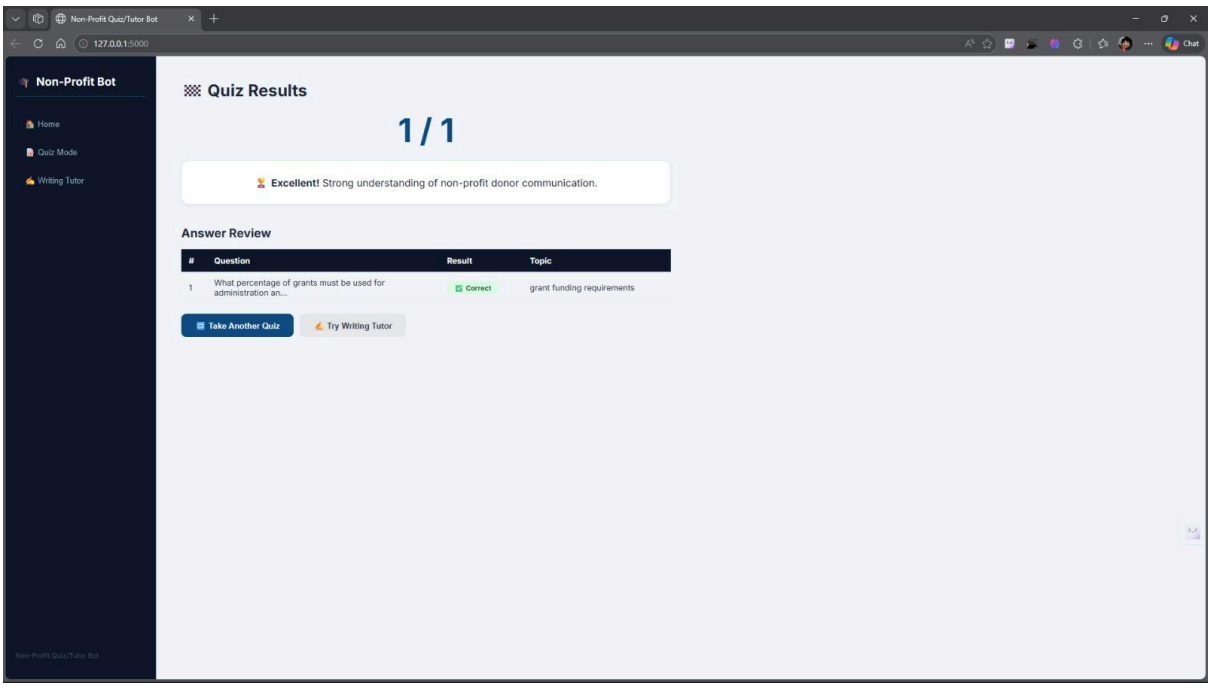
3. Results & Outputs

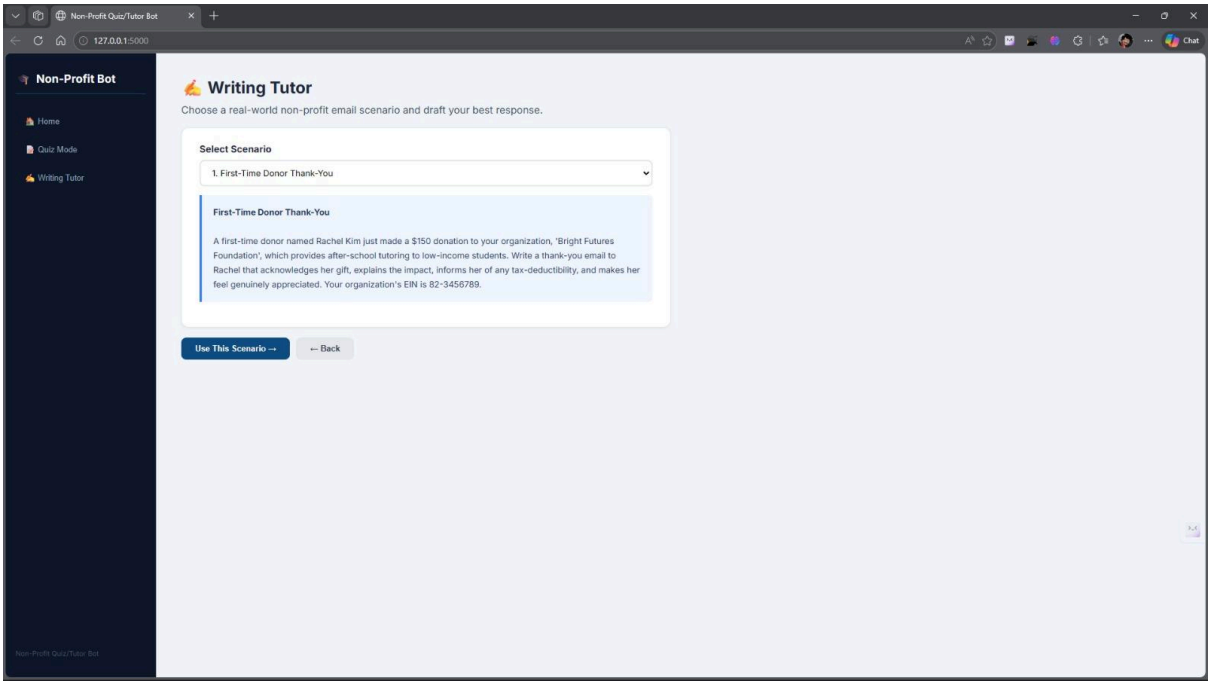
3.1. Screenshots

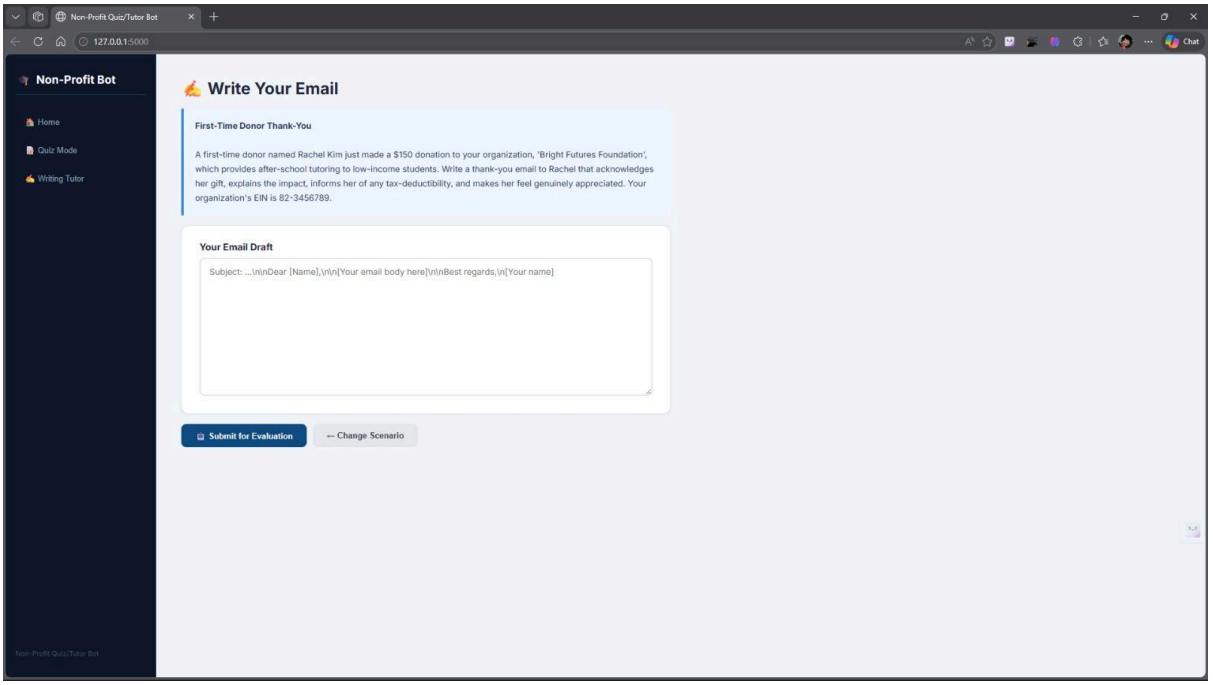





Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Q1 of 1
grant funding requirements · EASY
What percentage of grants must be used for administration and fundraising according to the Better Business Bureau's Wise Giving Alliance standard?
A. 45%
B. 35%
C. 25%
D. 65%
Show hint
Consider the standards that organizations like your donor may follow
Submit Answer


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Quiz Results
1 / 1
Excellent! Strong understanding of non-profit donor communication.
Answer Review
Question Result Topic
1 What percentage of grants must be used for administration an... Correct grant funding requirements
Take Another Quiz
Try Writing Tutor


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Writing Tutor
Choose a real-world non-profit email scenario and draft your best response.
Select Scenario
1. First-Time Donor Thank-You
First-Time Donor Thank-You
A first-time donor named Rachel Kim just made a $150 donation to your organization, 'Bright Futures Foundation', which provides after-school tutoring to low-income students. Write a thank-you email to Rachel that acknowledges her gift, explains the impact, informs her of any tax-deductibility, and makes her feel genuinely appreciated. Your organization's EIN is 82-3456789.
Use This Scenario →
← Back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Non-Profit Bot
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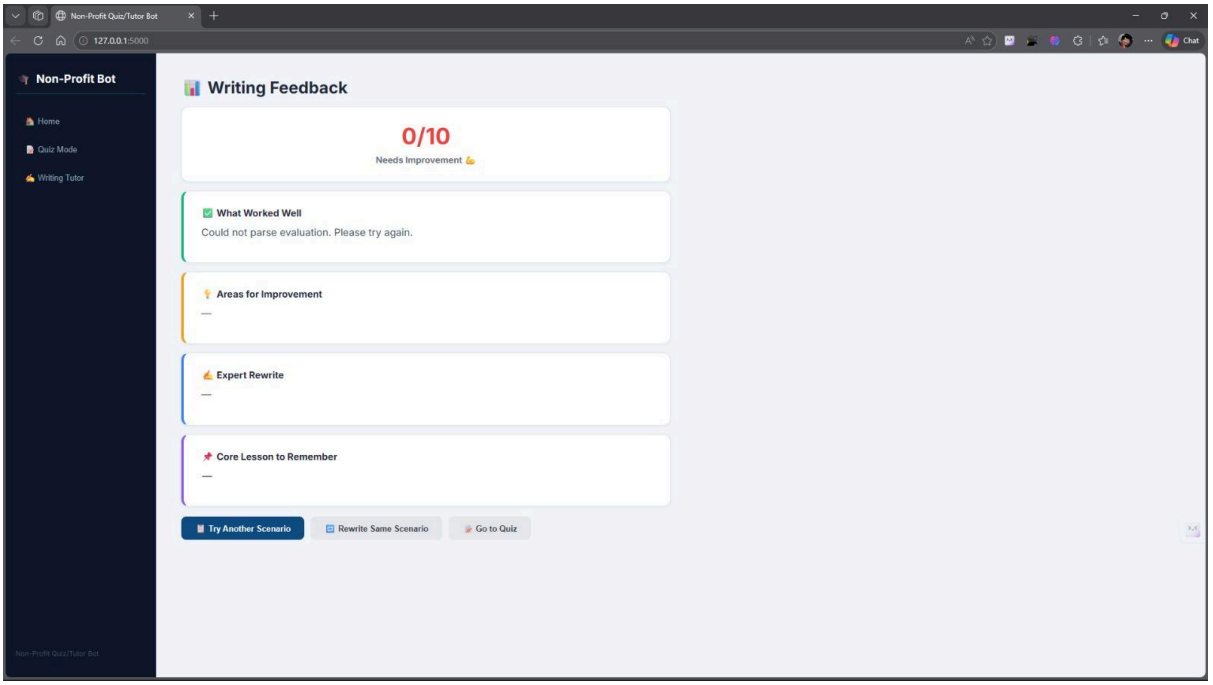
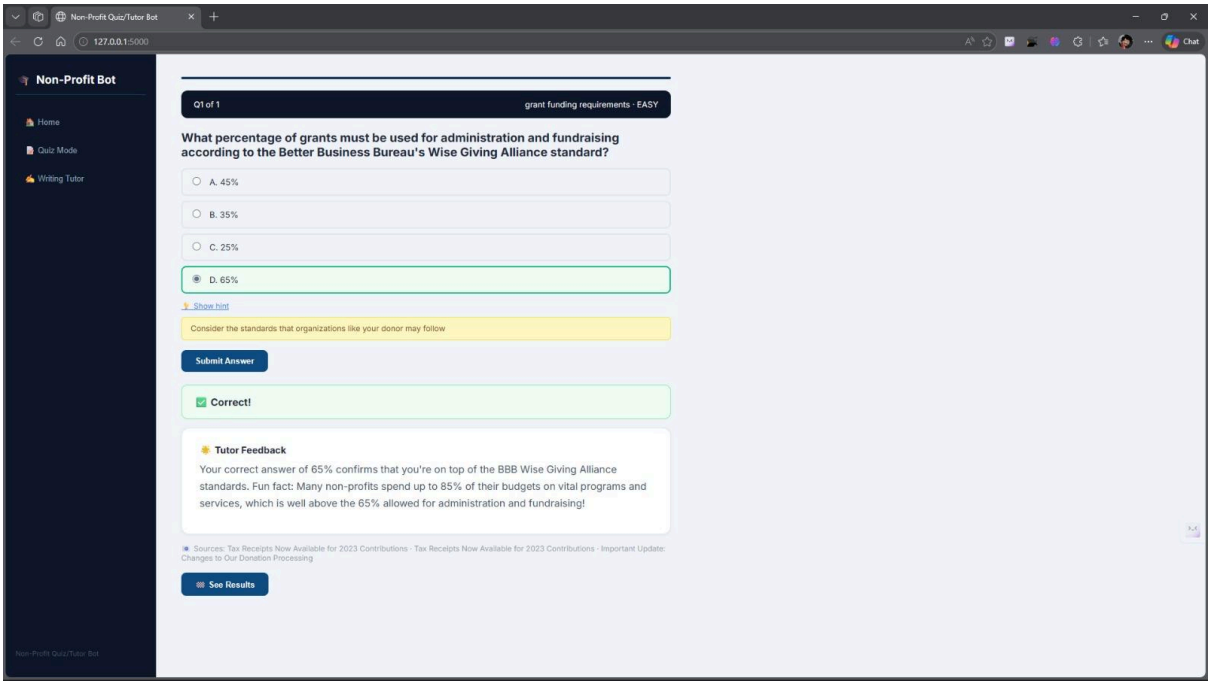
Quiz Mode
Writing Tutor
Write Your Email
First-Time Donor Thank-You
A first-time donor named Rachel Kim just made a $150 donation to your organization, 'Bright Futures Foundation', which provides after-school tutoring to low-income students. Write a thank-you email to Rachel that acknowledges her gift, explains the impact, informs her of any tax-deductibility, and makes her feel genuinely appreciated. Your organization's EIN is 82-3456789.
Your Email Draft
Subject: ...\n\nDear [Name],\n\n[Your email body here]\n\nBest regards,\n[Your name]
Submit for Evaluation
← Change Scenario


Non-Profit Quiz/Tutor Bot
127.0.0.1:5000
Non-Profit Bot
Home
Quiz Mode
Writing Tutor
Writing Feedback
0/10
Needs Improvement
What Worked Well
Could not parse evaluation. Please try again.
Areas for Improvement
—
Expert Rewrite
—
Core Lesson to Remember
—
Try Another Scenario
Rewrite Same Scenario
Go to Quiz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A. 45%
B. 35%
C. 25%
D. 65%
Show hint
Consider the standards that organizations like your donor may follow
Submit Answer
Correct!
Tutor Feedback
Your correct answer of 65% confirms that you're on top of the BBB Wise Giving Alliance standards. Fun fact: Many non-profits spend up to 85% of their budgets on vital programs and services, which is well above the 65% allowed for administration and fundraising!
Sources: Tax Receipts Now Available for 2023 Contributions · Tax Receipts Now Available for 2023 Contributions · Important Update: Changes to Our Donation Processing
See Results

3.2. Key Outcomes

- Developed an AI-driven interactive quiz/tutor bot for the non-profit domain
- Successfully integrated **LLM (LLaMA 3.1 via Groq API)** for intelligent evaluation and explanation
- Implemented **Retrieval-Augmented Generation (RAG)** for context-aware responses
- Built a vector database (ChromaDB) for efficient semantic search and knowledge retrieval
- Enabled automatic generation of quiz questions from donor email datasets
- Provided detailed and personalized feedback instead of simple right/wrong answers
- Created a modular and scalable system architecture
- Improved user engagement through a chatbot-based learning interface
- Achieved faster response time using in-memory caching
- Demonstrated practical application of Generative AI in education and non-profit training

4. Conclusion

The Non-Profit Quiz/Tutor Bot project successfully demonstrates the application of Generative AI in creating an interactive and intelligent learning platform. By integrating Large Language Models (LLaMA 3.1 via Groq API) with a Retrieval-Augmented Generation (RAG) approach and a vector database (ChromaDB), the system is able to generate meaningful quiz questions, evaluate user responses, and provide detailed contextual explanations. The modular architecture consisting of frontend, backend, AI, and data layers ensures smooth workflow, scalability, and efficient performance.

Throughout this project, we gained valuable experience in working with modern AI technologies, including LLM integration, vector databases, semantic search, and API-based system design. We also developed a strong understanding of building end-to-end applications, from data ingestion and processing to user interaction and feedback generation. Overall, the project highlights how AI can enhance learning experiences by making them more personalized, engaging, and effective, especially in specialized domains like the non-profit sector.

5. Future Scope & Enhancements

- User Authentication & Profiles: Implement login/signup and track user progress, scores, and learning history.
- Multi-User Support: Enable concurrent users with scalable cloud-based deployment.
- Advanced Personalization: Use AI to adapt question difficulty based on user performance.
- Real-Time Data Integration: Connect with live non-profit data sources and APIs for updated content.
- Improved UI/UX: Develop a more interactive and responsive interface (mobile app or advanced web UI).
- Multilingual Support: Support multiple languages to reach a wider audience.
- Voice-Based Interaction: Integrate speech-to-text and text-to-speech for voice-enabled learning.
- Enhanced Security: Add encryption, authentication, and secure API handling.
- LLM Optimization: Use fine-tuned models or multiple LLMs for better accuracy and reduced cost.
- Performance Optimization: Implement caching for LLM responses and asynchronous processing.